Cairo University Faculty Of Engineering	Final Project Statement
Biomedical Data Analytics	Grade: 15

Overview

For this project, each team is required to propose and build a real-world Medical Data Analytics application. This is not a notebook exercise or a script, it must be a fully functional application with a proper user interface that real users can interact with.

Core Requirements

Your application must fulfill the following three requirements:

1. **Collect Data from the User** The application must allow users (patients, doctors, or healthcare staff) to enter or upload their own medical data. This could include symptoms, lab results, health metrics, medical history, or any other clinically relevant input.
2. **Analyze the Data** The application must process and analyze the data provided by the user using appropriate data analytics and machine learning techniques to identify patterns, trends, or meaningful insights.
3. **Make a Prediction** Based on the analysis, the application must output a meaningful medical prediction or risk assessment — such as disease risk level, readmission likelihood, severity classification, or a health recommendation — presented clearly to the user.

What We Expect from Your Idea

When thinking about your project idea, your team should consider the following:

- It must be medically relevant and address a real healthcare problem.
- It must be practically useful, think like a developer building a tool that a real hospital, clinic, or patient could actually use.
- The application must have a real user interface (web or mobile), not just backend code.
- The machine learning model must be trained on a real, publicly available medical dataset.

Questions to Guide Your Thinking

Before finalizing your idea, your team should be able to answer these questions:

- Who is the user of your application? (Patient? Doctor? Hospital staff?)
- What data will the user input into your application?
- What analysis or ML model will your app use on that data?
- What prediction or result will your app show the user?
- How is this useful in a real clinical or healthcare setting?

Important Note: Submissions that are notebooks, scripts, or static analysis reports without a user-facing interface will not be accepted. Your deliverable is a working application, not an analysis report.